

Rishabh Dinesh Shukla

Spatial Data Specialist II | Data Engineer

✉ rishabhshukla445@gmail.com | 📞 +91 90825 74409 | 📍 Mumbai, India | 🛠️ rishabhshukla445-coder
| 🌱 rishabh-projects | 🌐 Portfolio

Results-driven **Data Engineer & Spatial Data Specialist** with 3+ years of experience at HERE Technologies designing and deploying production-grade ETL pipelines, web scraping infrastructure, and AI-powered automation systems. Proven track record of leading cross-functional teams, winning hackathons, and delivering scalable data solutions that eliminate thousands of engineering hours. Seeking to leverage deep expertise in Python, AWS, and geospatial data engineering within a high-impact engineering organization.

PROFESSIONAL EXPERIENCE

HERE Technologies

Jul 2022 – Present

Navi Mumbai, India

Spatial Data Specialist II

- Architected and deployed **Python-based ETL and web scraping pipelines** integrating internal proprietary systems with 20+ external REST APIs, reducing manual data acquisition time by approximately 60%.
- **Promoted to Subject Matter Expert (SME)** of Web Scraping Operations (WSO); directly managed, mentored, and quality-reviewed the output of a **15-member engineering team**, ensuring 100% on-time delivery across all sprint cycles.
- Engineered specialized web crawling solutions using **Scrapy**, custom middleware, and third-party scraping APIs to extract mission-critical geospatial datasets from over 50 complex, JavaScript-heavy websites.
- Developed and maintained **Streamlit-based operational dashboards** providing real-time visibility into scraping performance, error diagnostics, and data extraction accuracy, adopted by both engineering and QA teams.
- Applied **spatial data analysis and GIS methodologies** — including GeoJSON processing, PostGIS queries, and geospatial data validation — to ensure data integrity across location intelligence products used globally.
- Drove adoption of **data-driven decision-making** frameworks within the team, establishing KPIs for crawl efficiency, data consistency, and pipeline reliability, resulting in a measurable reduction in rework cycles.
- Partnered with cross-functional stakeholders to translate ambiguous business requirements into precise technical specifications, consistently delivering solutions ahead of schedule.
- **Hackathon Winner (1st Place — Efficiency Experts Category, HERE Mumbai AI InnoWave 2025):** Conceptualized, built, and deployed a **Generative AI-powered web scraping automation system** using Python, AWS Bedrock, Mistral LLM, Streamlit, and Boto3 — eliminating hardcoded spider maintenance and saving over **10,000+ engineering hours** annually.

TECHNICAL SKILLS

Languages & Query: Python (Advanced), SQL, Bash

Web Scraping & Crawling: Scrapy Framework, Selenium, BeautifulSoup, Web Scraping APIs, REST API Integration

AI / ML & GenAI: Large Language Models (LLMs), AWS Bedrock, Mistral LLM, Generative AI Workflows, AI-Prompted Automation

Cloud & Infrastructure: Amazon Web Services (AWS), Amazon S3, Boto3, Serverless Pipelines

Data & Visualization: Power BI, Streamlit, Pandas, Data Mining, ETL Pipeline Design, GeoJSON, PostGIS / PostgreSQL

GIS & Spatial: GIS Software, Spatial Databases, Geospatial Data Analysis, Location Intelligence

DevOps & Collaboration: Git, GitLab, JIRA, Agile / Scrum, Microsoft Excel, Microsoft PowerPoint

Leadership: Team Management (15+ members), Subject Matter Expertise, Stakeholder Communication, Operations Management

PROJECTS

Generative AI-Powered Scraper Automation

Aug 2025 – Sep 2025

Tech Stack: Python, AWS Bedrock, Mistral LLM, Streamlit, Boto3, Amazon S3

- Designed and delivered an end-to-end **GenAI-driven web scraping automation platform** that dynamically generates extraction logic via LLM prompting, completely replacing manually coded per-site spiders.
- Reduced site-level spider scripting time **from ~3 hours to near-zero** by leveraging Mistral LLM on AWS Bedrock to auto-generate JSON API extraction patterns from minimal input.
- Achieved estimated savings of **10,000+ engineering hours** per year; won **1st place (Efficiency Experts)** at HERE Mumbai AI InnoWave 2025 Hackathon.
- Built a Streamlit-based self-serve interface enabling non-technical stakeholders to trigger and monitor scraping jobs without engineering intervention.

AI Product Authenticity Analyzer — Agent Unveil*Tech Stack: Python, Computer Vision, Automated Scraping, LLM, Compliance Reporting*

- Built an **AI-powered product authenticity verification system** that cross-references product images and textual descriptions to detect misleading, fraudulent, or non-compliant e-commerce listings.
- Integrated **automated scraping workflows** with multi-modal comparison logic to flag inconsistencies between advertised product attributes and verified specifications.
- Generated structured **compliance reports** with confidence scores, significantly improving transparency and auditability for e-commerce marketplace operators.

Jun 2024 – Jul 2024

Single Spider Investigation Dashboard*Tech Stack: Python, Streamlit, Data Visualization, Web Scraping Diagnostics*

- Developed a **Streamlit analytics application** providing in-depth performance profiling of individual web scraping spiders, including crawl efficiency metrics, error log aggregation, and data extraction accuracy scores.
- Enabled both engineering and QA teams to make faster, data-driven decisions; incorporated features including **customizable charts, data export, and real-time monitoring** of active crawl jobs.
- Reduced mean time-to-resolution for scraping failures by surfacing root-cause diagnostics in a single unified dashboard, previously requiring manual log inspection.

Aug 2021 – Feb 2022

FOURNETRA — Assistive Spectacles for Drowsy, Comatose & Paralysed Individuals*Tech Stack: IoT, Embedded Systems, Assistive Technology, Morse Code Integration*

- Conceived and prototyped **smart assistive spectacles** tailored for four distinct user groups: individuals with sleep disorders, comatose patients, paralyzed patients, and Morse code users.
- Designed customized hardware-software interaction models to enhance communication, comfort, and quality of life for mobility-impaired individuals.
- Earned **Consolation Rank and Merit Placement** at the 16th AVISHKAR Research Convention 2021–2022 at both university and zonal levels.

CERTIFICATIONS

- **Career Essentials in Generative AI** — *Microsoft & LinkedIn Learning*
- **Data Analytics Essentials** — *Cisco Networking Academy*
- **Google-Endorsed Data & Digital Skills** — *Guvi, in collaboration with Google & Ministry of Skill Development, India*
- **Project Management Foundations** — *NASBA / Project Management Institute (PMI) via LinkedIn Learning*
- **Employability Skills & Placements Training** — *TechnoServe (2022)*
- **Certificate in MSc-IT Fundamentals** — *Comp Life Institute (2017)*

ACHIEVEMENTS & AWARDS

- **1st Place — Efficiency Experts Category**, HERE Mumbai AI InnoWave 2025 Hackathon; recognized for building a production-grade GenAI scraping automation system that saved 10,000+ engineering hours.
- **Consolation Rank & Merit Placement**, 16th AVISHKAR Research Convention 2021–2022 (University Level) for the FOURNETRA assistive spectacles project.
- **Merit Placement**, 16th AVISHKAR Research Convention 2021–2022 (Zonal Level).
- **Merit Placement**, 14th AVISHKAR Research Convention 2019–2020 (Zonal Level).

EDUCATION

Bachelor of Science in Information Technology

2019 – 2022

Model College, Mumbai University

Higher Secondary School Certificate (HSC)

2017 – 2019

Guardian High School and Junior College, Dombivli

Secondary School Certificate (SSC)

Completed 2017

New Sunrise English High School, Dombivli